

The Derivative as a Limit and as a Rate

Directions:

- An *average* rate over an interval is a slope between two points: $\frac{f(b) - f(a)}{b - a}$. An *instantaneous* rate at a point is what that slope becomes as the second point slides in: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$.
- Where a problem says “by the limit definition”, write the difference quotient, simplify until the h in the denominator cancels, and only then let $h \rightarrow 0$.
- Keep the units of a rate attached: output units per input unit.

1. The position of a cart on a track is $s(t) = t^2 + 3t$ metres at time t seconds. The table shows four of its positions.

t (s)	1	2	3	4
$s(t)$ (m)	4	10	18	28

Find the cart’s **average** velocity from $t = 1$ to $t = 3$ seconds.

Answer: _____ m/s

2. Let $f(x) = x^2$. Use the limit definition

$$f'(3) = \lim_{h \rightarrow 0} \frac{f(3 + h) - f(3)}{h}$$

to find the slope of the tangent line to $y = x^2$ at $x = 3$. Expand, cancel the h , then let $h \rightarrow 0$.

Answer: _____

- 3.** Same cart as problem 1: $s(t) = t^2 + 3t$ metres. Now find the **instantaneous** velocity at $t = 3$ seconds by evaluating

$$\lim_{h \rightarrow 0} \frac{s(3+h) - s(3)}{h}.$$

Compare your answer with problem 1 — one of them is a slope between two points on the curve, the other is the slope of the curve itself.

Answer: _____ m/s

- 4.** Let $f(x) = \frac{1}{x}$. Use the limit definition to find $f'(2)$. (Combine the two fractions in the numerator over a common denominator before you divide by h .)

Answer: _____

- 5.** A tank is draining. Its volume is $V(t) = 60 - 2t^2$ litres at time t minutes, for $0 \leq t \leq 5$.

(a) Use the limit definition to find $V'(5)$: _____

(b) Explain what that number says about the tank at $t = 5$ minutes. Give its units and say what the sign means.

6. Let $f(x) = \sqrt{x}$. Use the limit definition to find $f'(9)$. (Multiply the difference quotient by the conjugate of its numerator.) Give the exact value.

Answer: _____

7. Let $g(x) = x^3 - 2x$.

(a) Find the derivative function $g'(x)$: _____

(b) Use it to find the slope of the tangent line to $y = g(x)$ at $x = 2$: _____

8. Challenge. A ball is thrown straight up. Its height is $h(t) = 40t - 5t^2$ metres at time t seconds.

(a) Find the velocity function $h'(t)$: _____

(b) Find the velocity at $t = 2$ seconds, in metres per second: _____

(c) Compute the velocity at $t = 5$ seconds and explain, in one sentence, what it says about the ball at that instant.